

SIG31 : Task 7HD

Data Mining Challenge

1 Introduction

Tasks 5–8 are not obligatory; you can submit them in any order (or decide not to tackle them at all). C/D/HD is merely a subjective estimate of their difficulty level. For each task that you successfully complete, you score 10 points (and for those that are not 100% correct, no points will be given).

The due date for submission is **Week 6 (Sunday)**. After this due day, the evaluation team will not provide any feedback for your work in Olympus. The submission due day is **9th February (Sunday)**. Start tackling this task as early as possible. If we find your first solution incomplete or otherwise incorrect, you will still be able to amend it based on the generous feedback we will give you (allow 3–5 working days). In case of any problems/questions, do not hesitate to attend our on-campus/online classes or use the Discussion Board on the unit site.

Submitting after the aforementioned due date might incur a late penalty. The **cut-off date is Week 7 (Sunday)**. There will be **no extensions** and no solutions will be accepted thereafter. At that time, if your submission is not 100% complete, it will be marked as FAIL, without the possibility of correcting and resubmitting. To ensure a fair environment for all, we are always very strict about deadlines.

2 Task

Create a single Jupyter/IPython notebook (see the *Artefacts* section below for all the requirements), where you perform what follows.

1. Download at least five different datasets that are part of the NHANES study; see <https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?Cycle=2021-2023> and <https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?Cycle=2017-2020>. Merge them into a single data frame.
2. Using the `bokeh` package, which you will have to learn yourself (this is part of this HD-level task), create at least five nontrivial *interactive* data visualisations and/or tables.
3. Draw insightful and interesting conclusions. Do not forget to reflect on the potential data privacy and ethics issues that arise during the data analysis process.

Make it aesthetic and interesting to read.

*This HD-level task is purposely under-defined – you will not be told precisely what to do. Your aim is to discover, visualise, and explain some **interesting** relationships between **many** data features.*

In the course of the report preparation, you should apply a wide range of data frame wrangling techniques, including filtering, aggregation in groups, missing value handling, column transformation, etc. Do not use pie charts (as we discussed during the lecture). Please go beyond simple bar/box plots and histograms. The charts/tables must be interactive; a reader should be able to use slider bars, text boxes, etc. to control the data/view.

For some inspiration, you might want to take a look at some research papers citing the NHANES study: https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=nhanes.

3 Artefacts

The solution to the task must be included in a single Jupyter/IPython notebook (an .ipynb file) running against a Python 3 kernel. The use of Google Colab is discouraged. Nothing beats a locally-installed version where you have full control over the environment. Do not become dependent on third-party middlemen/distributors. Choose freedom instead.

Make sure that your notebook has a **readable structure**; in particular, that it is divided into sections. Use rich Markdown formatting (text in dedicated Markdown chunks – not just Python comments).

Do not include the questions/tasks from the task specification. Your notebook should read nicely and smoothly – like a report from data analysis that you designed yourself. Make the flow read natural (e.g., *First, let us load the data on... Then, let us determine... etc.*). Imagine it is a piece of work that you would like to show to your manager or clients — you certainly want to make a good impression. Check your spelling and grammar. Also, use formal language.

At the start of the notebook, you need to provide: the **title** of the report (e.g., *Task42:How Much I Love This Unit*), your **name, student number and email address**.

Then, add 1–2 introductory paragraphs (an introduction/abstract – what the task is about).

Before each nontrivial code chunk, briefly **explain** what its purpose is. After each code chunk, **summarise and discuss the obtained results** (in a few sentences).

Conclude the report with 1–2 paragraphs (summary/discussion/possible extensions of the analysis etc.).

Limitations of the ipynb-to-pdf renderer:

Ensure that your report as seen in Olympus is aesthetic. The ipynb-to-pdf renderer is imperfect. We work with what we have. Here are the most common Markdown-related errors.

- Do not include any externally loaded images (via the `! [label] (href)` Markdown command), for they lead to upload errors.
- Do not input HTML code in Markdown.
- Make sure you leave one blank line before and after each paragraph and bullet list. Do not use backslashes at the end of the line.
- Currently, also LaTeX formulae and Markdown tables are not recognised. However, they do not lead to any errors.

Checklist:

1. Header, introduction, conclusion (Markdown chunks).
2. Text divided into sections, all major code chunks commented and discussed in your own words (Markdown chunks).
3. Every subtask addressed/solved. In particular, all reference results that are part of the task specification have been reproduced (plots, computed aggregates, etc.).
4. The report is readable and neat. In particular:
 - all code lines are visible in their entirety (they are not too long),
 - code chunks use consecutive numbering (select *Kernel - Restart and Run All* from the Jupyter menu),

- rich Markdown formatting is used (# Section Title, * bullet list, 1. enumerated list, | table |, *italic*, etc.),
- the printing of unnecessary/intermediate objects is minimised (focus on reporting the results specifically requested in the task specification).

Submissions which do not *fully* (100%) conform to the task specification *on* the cut-off date will be marked as FAIL.

Good luck!

4 Intended Learning Outcomes

ULO	Is Related?
ULO1 (Data Processing/Wrangling)	YES
ULO2 (Data Discovery/Extraction)	YES
ULO3 (Requirement Analysis/Data Sources)	YES
ULO4 (Exploratory Data Analysis)	YES
ULO5 (Data Privacy and Ethics)	YES